

Each vial of Azithromycin for Injection 500 mg/vial contains 512.029 mg of Azithromycin (monohydrate) equivalent to 500.00 mg of Azithromycin (anhydrous).

PRODUCT DESCRIPTION

Azithromycin for Injection 500 mg/vial is a white lyophilized cake. The reconstituted solution is a clear colorless solution free from visible particles.

PHARMACOLOGY

The mode of action of azithromycin is inhibition of protein synthesis in bacteria by binding to the 50S ribosomal subunit and preventing translocation of peptides.

Microbiology

Azithromycin demonstrates activity in vitro against a wide range of bacteria including:

Gram-Positive Aerobic Bacteria: *Staphylococcus aureus*, *Streptococcus pyogenes* (group A β -haemolytic Streptococci), *Streptococcus pneumoniae*, α -haemolytic streptococci (viridans group) and other streptococci, and *Corynebacterium diphtheriae*. Azithromycin demonstrates cross-resistance with erythromycin-resistant gram-positive strains, including *Streptococcus faecalis* (enterococcus) and most strains of methicillin-resistant *Staphylococcus*.

Gram-Negative Aerobic Bacteria: *Haemophilus influenzae*, *Haemophilus parainfluenzae*, *Moraxella catarrhalis*, *Acinetobacter* sp, *Yersinia* sp, *Legionella pneumophila*, *Borderella pertussis*, *Borderella parapertussis*, *Shigella* sp, *Pasteurella* sp, *Vibrio cholerae* and *parahaemolyticus*, *Plesiomonas shigelloides*. Activities against *Escherichia coli*, *Salmonella enteritidis*, *Salmonella typhi*, *Enterobacter* sp, *Aeromonas hydrophila* and *Klebsiella* sp are variable and susceptibility tests should be performed.

Proteus sp, *Serratia* sp, *Morganella* sp and *Pseudomonas aeruginosa* are usually resistant.

Anaerobic Bacteria: *Bacteroides fragilis* and *Bacteroides* sp, *Clostridium perfringens*, *Peptococcus* sp and *Peptostreptococcus* sp, *Fusobacterium necrophorum* and *Propionibacterium acnes*.

Organisms of Sexually-Transmitted Diseases: Azithromycin is active against *Chlamydia trachomatis* and also shows good activity against *Treponema pallidum*, *Neisseria gonorrhoeae* and *Haemophilus ducreyi*.

Other organisms: *Borrelia burgdorferi* (Lyme disease agent), *Chlamydia pneumoniae*, *Mycoplasma pneumoniae*, *Mycoplasma hominis*, *Ureaplasma urealyticum*, *Campylobacter* sp and *Listeria monocytogenes*.

Opportunistic pathogens associated with HIV infections: *Mycobacterium avium-intracellulare*, *Pneumocystis carinii* and *Toxoplasma gondii*.

formed by N- and O-demethylation, by hydroxylation of the desosamine and aglycone rings and by cleavage of the cladinose conjugate. Comparison of HPLC and microbiological assays in tissues suggests that metabolites play no part in the microbiological activity of azithromycin.

Special Patient Groups: Elderly: In elderly (>65 years), slightly higher AUC values were seen after a 5-day regimen than in young patients (<40 years), but these are not considered clinically significant, and hence, no dose adjustment is recommended.

Renal Impairment: Following a single dose of azithromycin 1 g orally, the pharmacokinetics in subjects with mild to moderate renal impairment (GFR 10-80 mL/min) were not affected. Statistically significant differences in AUC₀₋₁₂₀ (8.8 mcg·hr/mL vs 11.7mcg·hr/mL), C_{max} (1 mcg/mL vs 1.6 mcg/mL) and ClCr (2.3 mL/min/kg vs 0.2 mL/min/kg) were observed between the group with severe renal impairment (GFR <10 mL/min) and the group with normal renal function.

Hepatic Impairment: In patients with mild (Class A) 10 moderate (Class B) hepatic impairment, there is no evidence of a marked change in serum pharmacokinetics of azithromycin compared to those with normal hepatic function. In these patients, urinary clearance of azithromycin appears to increase, perhaps to compensate for reduced hepatic clearance.

INDICATIONS

Treatment of community-acquired pneumonia caused by susceptible organisms including *Legionella pneumophila*, in patients who require initial IV therapy.

Treatment of pelvic inflammatory disease caused by susceptible organisms (*Chlamydia trachomatis*, *Neisseria gonorrhea*, *Mycoplasma hominis*), in patients who require initial IV therapy.

RECOMMENDED DOSE

Adults: Treatment of community acquired pneumonia due to the indicated organisms:
Recommended Dose:
500 mg as single daily dose by the IV route for at least 2 days. IV therapy should be followed by azithromycin by the oral route as a single daily dose of 500 mg to complete a 7-10 day course of therapy. The timing of the conversion to oral therapy should be done at the discretion of the physician and in accordance with clinical response.

Treatment of pelvic inflammatory disease due to the indicated organisms:
Recommended Dose:
500 mg as a single daily dose by the IV route for 1-2 days. IV therapy should be followed by azithromycin by the oral route at a single daily dose of 250 mg to complete a 7-day course of therapy. The timing of the conversion to oral therapy should be done at the discretion of the physician and in accordance with clinical response. If anaerobic microorganism are suspected of contributing to the infection, an antimicrobial anaerobic agent may be administered in combination with azithromycin.

Elderly
The same dosage as in adult patients is used in the elderly.

Patients with Renal Impairment
No dosage adjustment is necessary in patients with mild to moderate (GFR 10-80 mL/min) or severe (GFR <10 mL/min) renal impairment.

Patients with Hepatic Impairment

The same dosage as in patients with normal hepatic function may be used in

patients with mild to moderate hepatic impairment.

Children

The safety and efficacy of IV azithromycin for the treatment in children have not been established.

MODE OF ADMINISTRATION

Administration

After reconstitution and dilution, the recommended route of administration for IV azithromycin is by IV infusion only. It is not recommended for administration as an IV bolus or an IM injection.

The infusate concentration and rate of infusion of azithromycin IV should be either 1 mg/mL over 3 hrs or 2 mg/mL over 1 hr. An IV dose of 500 mg should be infused for a minimum duration of 1 hr.

Reconstitution: Prepare the initial IV solution for infusion by adding 4.8 mL of sterilised water for injection to the 500-mg vial and shaking the vial until all of the drug is dissolved. Since azithromycin IV is supplied under vacuum, it is recommended that a standard 5 mL (non-automated) syringe be used to ensure that the exact amount of 4.8 mL of sterilised water for injections is dispensed. Each mL of reconstituted solution contains 100-mg azithromycin.

SYMPTOMS AND TREATMENT OF OVERDOSE

Adverse events experienced in higher than recommended doses were similar to those seen at normal doses. In the event of overdosage, general symptomatic and supportive measures are indicated as required.

CONTRAINDICATIONS

Patients with a history of allergic reactions to azithromycin or any of the macrolide antibiotics.

WARNINGS AND PRECAUTIONS

in the event of severe acute hypersensitivity reactions, such as anaphylaxis, severe cutaneous adverse reactions (SCARs) [e.g. Stevens-Johnson Syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis

(AGEP), AZITHROMYCIN MYLAN 500 mg Injection should be discontinued immediately and appropriate treatment should be urgently initiated.

As with erythromycin and other macrolides, rare serious allergic reactions including angioedema and anaphylaxis (rarely fatal), have been reported. Some of these reactions with azithromycin have resulted in recurrent symptoms and required a longer period of observation and treatment.

Since the liver is the principal route of elimination for azithromycin, the use of azithromycin should be undertaken with caution in patients with significant hepatic disease.

In patients receiving ergot derivatives, ergotism has been precipitated by co-administration of some macrolide antibiotics. There are no data concerning the possibility of an interaction between ergot and azithromycin. However, because of the theoretical possibility of ergotism, azithromycin and ergot derivatives should not be co-administered.

As with any antibiotic preparation, observation for signs of superinfection with nonsusceptible organisms, including fungi, is recommended. In patients with severe renal impairment (GFR <10 mL/min), a 33% increase in systemic exposure to azithromycin was observed.

Prolonged cardiac repolarization and at interval, imparting a risk of developing cardiac arrhythmia and torsade de pointes, have been seen in treatment with other macrolides. A similar effect with azithromycin cannot be completely ruled out in patients at increased risk for prolonged cardiac repolarization

Hypersensitivity

As with erythromycin and other macrolides, rare serious allergic reactions, including angioedema and anaphylaxis (rarely fatal), dermatologic reactions including Stevens - Johnson syndrome (SJS), Toxic Epidermal Necrolysis (TEN) (rarely fatal), and Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) have been reported. Some of these reactions with azithromycin have resulted in recurrent symptoms and required a longer period of observation and treatment.

If an allergic reaction occurs, the drug should be discontinued and appropriate therapy should be instituted. Physicians should be aware that reappearance of the allergic symptoms may occur when symptomatic therapy is discontinued.

Prolongation of the QT interval

Prolonged cardiac repolarization and QT interval, imparting a risk of developing cardiac arrhythmia and torsade's de pointes, have been seen in treatment with macrolides, including azithromycin. Prescribers should consider the risk of QT prolongation, which can be fatal, when weighing the risks and benefits of azithromycin for at-risk groups including:

- Patients with congenital or documented QT prolongation
- Patients currently receiving treatment with other active substances known to prolong QT interval, such as antiarrhythmics of Classes IA and III, antipsychotic agents, antidepressants, and fluoroquinolones.
- Patients with electrolyte disturbance, particularly in cases of hypokalaemia and hypomagnesaemia
- Patients with clinically relevant bradycardia, cardiac arrhythmia or cardiac insufficiency
- Elderly patients: elderly patients may be more susceptible to drug-associated effects on the QT interval

Infantile hypertrophic pyloric stenosis (IHPS) has been reported following the use of azithromycin in infants (treatment up to 42 days of life). Parents and caregivers should be informed to contact their physician if vomiting and/or irritability with feeding occurs.

Effects on the Ability to Drive or Operate Machinery

There is no evidence to suggest that azithromycin may have an effect on the patient's ability to drive or operate machinery.

PREGNANCY AND LACTATION

Animal reproductive studies have demonstrated that azithromycin crosses the placenta, but have revealed no evidence of harm to the foetus. There are no data on secretion in breast milk. Safety for use in human pregnancy and lactation has not been established.

Azithromycin should only be used in pregnant or lactating women where adequate alternatives are not available.

ADVERSE DRUG REACTIONS

Azithromycin is well tolerated with a low incidence of side effects. The following undesirable effects have been reported:

Blood and Lymphatic System Disorders:
Transient episodes of mild neutropenia have occasionally been observed, although a causal relationship to azithromycin has not been established.

Ear and Labyrinth Disorders:
Hearing impairment (including hearing loss, deafness and/or tinnitus) has been reported in some patients receiving azithromycin.

Gastrointestinal Disorders:
Nausea, vomiting, diarrhea, loose stools, abdominal discomfort (pain/cramps) and flatulence.

Hepatobiliary Disorders:
Abnormal liver function.
Skin and Subcutaneous Tissue Disorders:
Allergic reactions including rash and angioedema.

General Disorders and Administration Site Disorders:
Local pain and inflammation at the site of infusion.

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The following undesirable effects have been reported in association with DMAC prophylaxis and treatment:

Diarrhoea, abdominal pain, nausea, loose stools, flatulence, vomiting, dyspepsia, rash, pruritus, headache and arthralgia.

When azithromycin 600 mg is given daily for the treatment of DMAC infection for prolonged periods, the most frequently reported treatment-related side effects are abdominal pain, nausea, vomiting, diarrhoea, flatulence, headache, abnormal vision and hearing impairment.

The following additional undesirable effects have been reported:

Infections and Infestations:

Moniliasis and vaginitis.

Blood and Lymphatic System Disorders:

Thrombocytopenia.

Immune System Disorders:

Anaphylaxis (rarely fatal)

Metabolism and Nutrition Disorders:

Anorexia.

Psychiatric Disorders:

Aggressive reaction, nervousness, agitation and anxiety.

Nervous System Disorders:

Dizziness, convulsions (as seen with other macrolides), headache, hyperactivity, paresthesia, somnolence and syncope. There have been rare reports of taste perversion.

Ear and Labyrinth Disorders:

Vertigo.

Vascular Disorders:

Hypotension.

Gastrointestinal Disorders:

Vomiting/diarrhea (rarely resulting in dehydration), dyspepsia, constipation, pseudomembranous colitis, pancreatitis and rare reports of tongue discoloration.

Hepatobiliary Disorders:

Hepatitis and cholestatic jaundice have been reported, as well as rare cases of hepatic necrosis and hepatic failure, which have rarely resulted in death. However, a causal relationship has not been established.

Musculoskeletal and Connective Tissue Disorders:

Arthralgia.

Renal and Urinary Disorders:

Interstitial nephritis and acute renal failure.

General Disorders and Administration Site Conditions:

Asthenia has been reported, although a causal relationship has not been established; fatigue and malaise. Local pain and inflammation at the site of infusion.

Post-marketing experience:

Cardiac Disorders: Palpitations and arrhythmias including ventricular tachycardia have been reported. There have been rare reports of QT prolongation and torsades de pointes (see **Warnings and Precautions for Use**).

Skin and Subcutaneous Tissue Disorders

Frequency not known: severe cutaneous adverse reactions (SCARs) including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP).

Allergic reaction including pruritus, rash, photosensitivity, edema, urticarial, and angioedema. Rarely, serious cutaneous adverse reactions including erythema multiforme, SJS, TEN and DRESS have been reported.

Gastrointestinal Disorders:

Infantile hypertrophic pyloric stenosis.

INTERACTIONS WITH OTHER MEDICAMENTS

Zidovudine:

Single 1000 mg doses and multiple 1200 mg or 600 mg doses of azithromycin had little effect on the plasma pharmacokinetics or urinary excretion of zidovudine or its glucuronide metabolite. However, administration of azithromycin increased the concentrations of phosphorylated zidovudine, the clinically active metabolite, in peripheral blood mononuclear cells. The clinical significance of this binding is unclear, but it may be of benefit to patients.

Azithromycin does not interact significantly with the hepatic cytochrome P-450 system. It is not believed to undergo the pharmacokinetic drug interactions as seen with erythromycin and other macrolides. Hepatic cytochrome P-450 induction or inactivation via cytochrome-metabolite complex does not occur with azithromycin.

Ergot:

Due to the theoretical possibility of ergotism, the concurrent use of azithromycin with ergot derivatives is not recommended.

Pharmacokinetic studies have been conducted between azithromycin and the following drugs known to undergo significant cytochrome P-450 mediated metabolism.

Atorvastatin:

Co-administration of atorvastatin (10 mg daily) and azithromycin (500 mg daily) did not alter the plasma concentrations of atorvastatin (based on a HMG CoA-reductase inhibition assay).

Carbamazepine:

No significant effect was observed on the plasma levels of carbamazepine or its active metabolite in patients receiving concomitant azithromycin.

Cimetidine:

In effects of a single dose of cimetidine, given 2 hrs before azithromycin, on the pharmacokinetics of azithromycin, no alteration of azithromycin pharmacokinetics was seen.

Coumarin-Type Oral Anticoagulants:

Azithromycin did not alter the anticoagulant effect of a single 15-mg dose of warfarin administered to healthy volunteers. There have been reports received of potentiated anticoagulation subsequent to co-administration of azithromycin and coumarin-type oral anticoagulants. Although a causal relationship has not been established, consideration should be given to the frequency of monitoring prothrombin time when azithromycin is used in patients receiving coumarin-type oral anticoagulants.

Cyclosporin:

The resulting cyclosporin C_{max} and AUC₀₋₅ were found to be significantly elevated after administration of a 500-mg/day oral dose of azithromycin for 3 days followed by administration of a single 10-mg/kg oral dose of cyclosporine. Consequently, caution should be exercised before concurrent administration of these drugs. If co-administration is necessary, cyclosporin levels should be monitored and the dose adjusted accordingly.

Efavirenz:

Co-administration of a 600-mg single dose of azithromycin and 400 mg efavirenz daily for 7 days did not result in any clinically significant pharmacokinetic interactions.

Fluconazole:

Co-administration of a single dose of 1200 mg azithromycin did not alter the pharmacokinetics of a single dose of 800 mg fluconazole. Total exposure and half-life of azithromycin were unchanged by the co-administration of fluconazole; however, a clinically insignificant decrease in C_{max} (18%) of azithromycin was observed.

Indinavir:

Co-administration of a single dose of 1200 mg azithromycin had no statistically significant effect on the pharmacokinetics of indinavir administered as 800 mg 3 times daily for 5 days.

Methylprednisolone:

In a pharmacokinetic interaction study in healthy volunteers, azithromycin had no significant effect on the pharmacokinetics of methylprednisolone.

Midazolam:

In healthy volunteers, co-administration of azithromycin 500 mg/day for 3 days did not cause clinically significant changes in the pharmacokinetics and pharmacodynamics of a single 15-mg dose of midazolam.

Nelfinavir:

Co-administration of 1200 mg azithromycin and nelfinavir (750 mg 3 times daily to steady-state) produced no clinically significant effect. Dose adjustment is not necessary.

Rifabutin:

Co-administration of azithromycin and rifabutin did not affect the serum concentrations of either drug. Neutropenia was observed in subjects receiving concomitant treatment with azithromycin and rifabutin. Although neutropenia has been associated with the use of rifabutin, a causal relationship to combination with azithromycin has not been established.

Sildenafil:

There was no evidence of an effect of azithromycin (500 mg daily for 3 days) on the AUC and C_{max} of sildenafil or its major circulating metabolite.

Terfenadine:

No reports of an interaction between azithromycin and terfenadine. There have been rare cases reported where the possibility of such interaction could

be entirely excluded; however, there was no specific evidence that such an interaction had occurred.

Theophylline:

There is no evidence of any pharmacokinetic interaction when azithromycin. and theophylline are co-administered to healthy volunteers.

Triazolam:

Co-administration of azithromycin 500 mg on day 1 and 250 mg on day 2 with 0.125 mg triazolam on day 2 had no significant effect on any of the pharmacokinetic variables for triazolam.

Trimethoprim/Sulfamethoxazole:

Co-administration of trimethoprim/sulfamethoxazole DS (160 mg/800 mg) for 7 days with azithromycin 1200 mg on day 7 had no significant effect on peak concentrations, total exposure or urinary excretion of either trimethoprim or sulfamethoxazole.

Incompatibilities

Other IV substances, additives or medications should not be added to IV azithromycin, or infused simultaneously through the same IV line.

CAUTIONS FOR USAGE

Chemical and physical in-use stability of the reconstituted product has been demonstrated for 24 hrs at 25°C. When diluted according to the instructions the diluted solution is chemically and physically stable for 24 hrs at or below 30°C or for 7 days if stored under refrigeration at 2-8°C.

From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage and conditions prior to use are the responsibility of the user and would normally be no longer than 24 hrs at 2-8°C, unless reconstitution and dilution have taken place in controlled and validated aseptic conditions.

Dilute the solution further prior to administration as follows:

Dilution: To provide azithromycin over a concentration range of 1-2 mg/mL, transfer 5 mL of the 100 mg/mL azithromycin solution into the appropriate amount of any of the diluents as follows.

Final infusion solution concentration of 1 and 2 mg/mL requires 500 and 250 mL amount of diluent, respectively.

The reconstituted solution can be diluted with:

0.9% NaCl Injection, 5% Dextrose in 0.45% NaCl Injection with 20mEq KCl, 5% Dextrose in Lactated Ringers Injection, 5% Dextrose and 0.45% NaCl Injection, Normosol-R in 5% Dextrose Injection and Normosol-M in 5% Dextrose Injection to a concentration of 2.0mg/mL.

Parenteral drug products should be inspected visually for particulate matter prior to administration. If particulate matter is evident in reconstituted fluids, the drug solution should be discarded.

STORAGE

Store below 30°C. The diluted solution is stable up to 7 days at 2-8°C and 24 hours at room temperature below 30°C.

AVAILABILITY

500 mg Vial: Carton of 1x1 Vial, Carton of 1x10 vials

For further information, please consult your physician or pharmacist.

Date of Revision: 04-2019

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Manufactured by:
Mylan Laboratories Limited
(Sterile Product Division)
Opp. IIM, Bilekahalli, Bannerghatta Road,
Bangalore – 560 076,
INDIA.



BACK SIDE

ARTWORK DETAIL LABEL							
PRODUCT	Azithromycin for Injection - 500 mg						
BUYER / COUNTRY	Mylan / Malaysia						
DIMENSION	280 x 320 mm	Component	Pack Insert	Pack	-	No. of Colours	1
COLOUR SHADES	 Black						
VERSION & DATE	Ver: 13; Date: 15.04.2019						
SPECIAL INSTRUCTIONS	Dimensions may change during commercial supply.						